



# Evidence in Support of Analogical Reasoning Improvements with Executive Attention Intervention in Healthy Young Adults

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Received: 19 April 2022 / Accepted: 13 June 2022 / Published online: 19 August 2022

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**Abstract** Analogical reasoning improvement is important in educational outcome improvement. Inspired by recent ideas and evidence, we applied anti-saccade task training as an executive attention intervention and tested whether it could improve analogical reasoning performance. A serial-task paradigm was applied where participants performed an anti-saccade followed by an analogical reasoning task including a perception condition. The experimental group finished the anti-saccade task in which the ratio of anti-saccade trials to pro-saccade trials was 5:1 while the counterpart was 1:1 in the active control group. Also, a blank control group was established where participants merely finished the analogical reasoning task. Event-related electroencephalographic (EEG) data were recorded when participants were performing the executive attention and analogical reasoning tasks. In addition, their resting state EEG was collected before and after the executive attention intervention. Behaviorally, the experimental group reacted significantly faster than the other two groups in analogical reasoning but not in perception. At the neural level, in the

experimental group alone, the anti-saccade trials elicited a smaller N2 than pro-saccade trials and the resting alpha power was improved after executive attention intervention. No significant difference in P2 was found between the two groups in analogical reasoning or perception but the experimental group showed a larger late positive component than the active control group in analogical reasoning. We also found that the late positive component mediated the relationship between the N2 of anti-saccade trials and analogical reasoning reaction times in the experimental group. We further discussed the role of executive attention in the analogical reasoning process, which may pave the way for the future reliable improvement of fluid intelligence.

**Keywords** Analogical reasoning · Executive attention · Cognitive intervention · Anti-saccade task · Electroencephalography

## Introduction

Analogical reasoning, which refers to the ability to recognize relational similarities between disparate or dissimilar phenomena in different contexts [1], is thought to be a core cognitive skill that distinguishes humans from all other species [2, 3] and contributes to fluid intelligence [2, 4]. Previous studies have claimed that analogical reasoning is ubiquitous in math and science education [5], and thus analogical reasoning improvement is considered of great importance in educational outcome improvements. Although either bolstering the cognitive capacity of the reasoner or lessening their cognitive load can improve the analogical reasoning process [6], previous studies have mainly focused on the latter [7–9]. Thus, studies are needed to explore if and how a cognitive

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intervention focusing on training a certain domain-general ability can improve analogical reasoning performance.

Executive attention, a general ability to voluntarily control attention to regulate thoughts and behavior [10], is considered the cornerstone of fluid intelligence [11, 12] that guides cognitive integration in fluid intelligence [13]. Previous studies have also reported that fluid intelligence can be significantly improved *via* a series of training tasks related to executive attention [14]. Since analogical reasoning is a core component of fluid intelligence [15] that can be predicted in children using executive attention levels measured by the Stroop task [16], executive attention intervention may effectively improve analogical reasoning performance.

Among executive attention-related tasks, anti-saccade (AS) tasks [17] are considered one of the most reliable paradigms reflecting executive attention [18, 19]. Moreover, researchers have reported that AS tasks tend to dominate latent factors of attention compared to other executive attention-related tasks [20], indicating that AS task training can accurately and efficiently improve executive attention. Typically, the AS task consists of AS and pro-saccade (PS) conditions. It has been proposed that executive attention is required more in AS than in PS [18, 21]. In the AS condition, individuals actively inhibit distracting cues and shift attention to the other side after the cue. A previous study found that individuals with high cognitive capacity perform better only on the AS condition than low-capacity individuals [22], and a recent study reported that the mean amplitude of saccades occurring during a reasoning task (mean saccade amplitude) is larger when the problem is solved than when it is not [23]. These results implied that active inhibitory control and attentional shift are closely related to cognitive performance. Specifically, analogical reasoning problem-solving requires individuals to suppress distracting features and switch attention quickly and flexibly between different objects to extract and integrate the appropriate relationship [24]. Thus, if individuals improve inhibitory control and attentional shift in executive attention training *via* AS tasks, they may further improve analogical reasoning performance. On the contrary, if individuals do not develop a pattern of active inhibitory control and attentional shift in AS tasks, neither executive attention nor the performance of reasoning could be improved. To test this conjecture, we manipulated the ratio of AS trials to PS trials to establish a high executive attention group (HEA, 5:1) and a low executive attention group (LEA, 1:1). In the HEA group, participants had to frequently and actively shift attention in AS training, but in the LEA group, the reaction was passive as there were no predictable cues for AS or PS trials while they were equal in number and pseudo-random, which ensured that participants in the LEA group did not consciously develop a pattern of active inhibitory control and attentional shift. Namely, the LEA group, as the active control group, would

not have improved executive attention. We hypothesized that the HEA group would have enhanced executive attention, which may further enhance analogical reasoning, but this would not happen in the LEA group.

Furthermore, accumulating EEG data is helpful to provide valuable evidence of the impact of executive attention intervention at the level of brain function. N2, a negative component at ~200 ms after the onset of stimulation, is thought to reflect the attention process [25]. It has been found that during AS tasks, AS trials evoke a more negative N2 component than PS trials when the cue prior to the target is present, and it is regarded as an index reflecting the ability to inhibit the distracting stimulus and actively shift attention [26]. Moreover, the late positive component (LPC) may reflect a certain functional process in specific reasoning tasks [27–30]. For instance, Qiu *et al.* (2008) found that difficult analogical reasoning tasks evoke a larger P600–1000 effect than easy tasks in encoding, implying that stimuli-locked LPC reflects relation integration in analogical reasoning tasks. In addition to event-related potentials, the impact of executive attention intervention may be reflected by cortical arousal. It is well known that alpha-band activity is associated with attentional processing and helps resolve behavioral conflicts during congruency tasks [31–35]. Notably, a recent study found that a single session of attention training improves resting anterior alpha activity [36].

Recently, Tsukahara *et al.* [37] found that perceptual discrimination is also related to fluid intelligence performance, where executive attention plays an important role. It is necessary to determine whether such improved higher-order reasoning might be due to enhanced lower-level sensory ability [38]. Hence, we planned to set a perceptual discrimination task to test this possibility. If there was no influence of executive attention intervention on perception ability, it would provide a cutting-edge contrast for the influence of executive attention intervention on analogical reasoning performance. In the present study, we chose a typical analogical reasoning task called the relational matching-to-sample task [39] that involves a second-order comparison between the relations formed by two objects [40], which has been widely used as a test for analogical reasoning capabilities [41]. Importantly, a perception matching task was included to identify perception ability [13]. Meanwhile, since the ceiling effect on behavior may prevent us from revealing actual changes and even event-related potential (ERP) results are more sensitive, we also analyzed P2, which has been proposed as an index of perceptual discrimination [42, 43].

Finally, it has been suggested that serial-task paradigms can provide direct and supportive evidence for cognitive training (intervention) [44]. Thus, we applied a design with reference to a previous study [45] where participants completed executive attention tasks first and then performed analogical reasoning tasks at short intervals with EEG recorded

for ERP analysis. Besides, we also set a blank control group where a relatively large sample of participants was recruited from the same environment but required to only complete the analogical reasoning task without any influential pre-task or EEG recording.

Behaviorally, we predicted that the HEA group would show better performance in the AS condition than the LEA group. They would also perform better than the LEA group and the blank control group in the analogical reasoning task, but not in the perception task. At the neural level, N2 amplitude would be reduced due to the adaptation to interference [46] and an increased executive attention level [47], and cue-locked N2 amplitudes of AS trials would be smaller during the intervention only in the HEA group. In some recent studies, after cognitive training, positive ERP components became larger, indicating enhanced focusing of attention on task-related processing [48–50]. Hence, if the ability to concentrate on relation integration improved, the HEA group would show larger LPC amplitudes in the analogical reasoning task than the LEA group. Moreover, recent eye-tracking studies have found that to solve four-term analogies (A:B::C:D) or scene analogies successfully, individuals initially control their attention to the A:B pair [51, 52] and try to block distractors [24, 53], suggesting that executive attention contributes to relation integration, which then helps solve analogical reasoning problems correctly and efficiently [54]. Thus, we hypothesized that relation integration may mediate the relationship between executive attention interventions and analogical reasoning performance. We established a structural equation model and incorporated cue-locked N2 amplitudes of the AS trials, stimulus-locked LPC amplitudes of the analogical reasoning task, and analogical reasoning behavioral performance to test this hypothesis. In the present study, we aimed to provide valuable information by obtaining insightful cognitive and neural evidence.

## Materials and Methods

### Participants

An *a priori* sample size calculation using G\*Power v. 3.1 software [55] indicated that a total of at least 54 participants would be required for a  $2 \times 2$  mixed-design repeated-measures analysis of variance (ANOVA), including an interaction, for a moderate effect size ( $f=0.25$ ), at standard alpha and power values ( $1-\beta$ ) (0.05 and 0.95 respectively). In all, we recruited 73 healthy volunteers randomly assigned to either the HEA group (37 participants; 13 males; age:  $19.89 \pm 1.54$  years) or the LEA group (36 participants; 14 males; age:  $20.03 \pm 1.25$  years). The groups did not differ in sex or age ( $P>0.6$ ). In addition, we recruited 100 healthy

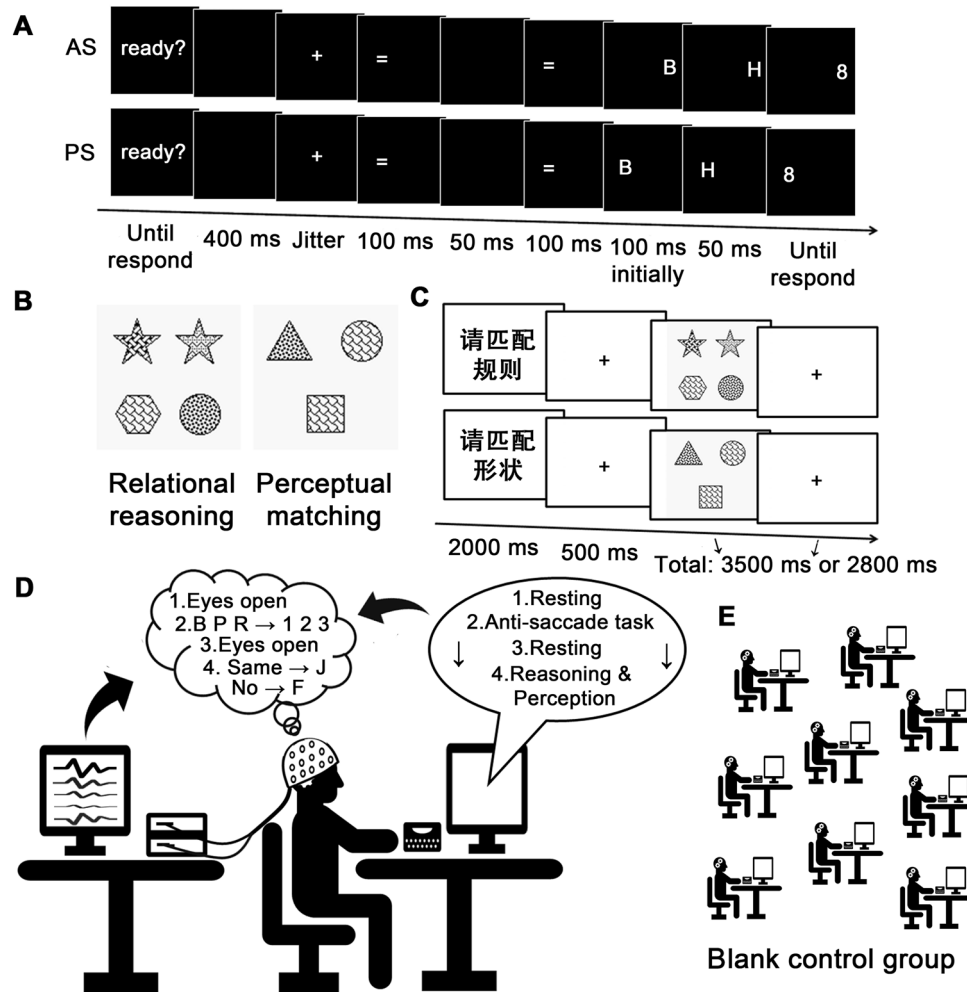
volunteers (34 males; age:  $20.18 \pm 1.49$  years) as the blank control group. All participants were right-handed, had normal or corrected-to-normal vision, and reported no history of nervous system disease. All participants provided handwritten informed consent and were naive to the experimental purpose throughout the experiment. The participants were given appropriate reimbursement at completion. The study was approved by Southwest University Human Ethics Committee for Human Research.

### Apparatus and Tasks

Participants performed experiments in a soundproof room. The experimental stimuli were presented on a 19-inch color monitor (85 Hz refresh rate, 1024 pixels  $\times$  768 pixels). Stimulus presentation and data acquisition were accomplished by programs designed with E-prime 2.0 (Psychology Software Tools, Inc., Pittsburgh, USA).

#### *Anti-saccade Task*

In this task, each session began with the presentation of a yellow character “ready?” at the center of the screen. The character disappeared once the participant indicated their readiness by pressing the keyboard space bar. In each trial, a 400-ms blank screen first appeared, followed by the fixation symbol “+” at the center of the screen. The duration of fixation alternated between 200 ms and 2200 ms in 400-ms increments. Immediately after fixation, an “=” cue appeared at  $\sim 12^\circ$  of the participant’s visual angle to the left or right of the fixation for 100 ms. Then, a second cue was presented for 100 ms following a 50-ms blank screen in the same eccentric location. After that, the target letters (B, P, or R) appeared at the same or the opposite location as the cue, depending on the trial type. Each letter corresponded to a response button (index, middle, or ring finger of the right hand for “1”, “2”, and “3” respectively). The participants were asked to respond to the target as quickly and accurately as possible. The target was followed by a succession of backward-masking stimuli: an “H” for 50 ms, then the number “8”. The “8” would remain until the participant made a response. The background was always black across the task. Target duration was manipulated from one session to another to keep the training adaptive. For participants in both the HEA and LEA groups, once the average accuracy rate reached 60% in a certain session, the duration would be shortened by 10 ms in the next session. If not, the duration was increased by 10 ms (the initial duration was set as 100 ms). Note that the duration was not shorter than 30 ms, so if the duration was reduced to 30 ms in one block, the duration of the next block was kept at 30 ms, even if the average accuracy rate was  $>60\%$ . This task was segmented into five blocks of 144 trials each. This included AS trials and PS trials (Fig. 1A) that were mixed



**Fig. 1** Experimental details. **A** The sequences of anti-saccade (AS) and pro-saccade (PS) trials in the AS task, in which the duration of fixation is altered from 200 ms to 2200 ms in 400-ms increments. The font color of “ready” was yellow. **B** Examples of stimuli for the relational matching-to-sample task. In the relational reasoning, participants indicate the dimension of change between the top pair of objects (e.g., texture or shape) and then determine whether the bottom pair of objects changes along the same dimension. In the perceptual matching, participants indicate whether the bottom object

matches either one of the top two objects along the specified dimension, according to a preceded cue (it is “shape” in the above example). **C** Trial sequences in the relational reasoning task (upper) and the perceptual matching task (lower). The Chinese introductions refer to the matching change rule (upper) and matching shape rule (lower). Here, the correct answers to the examples are “yes” (reasoning task) and “no” (perceptual task). **D** The design and procedure of this study. **E** Collection of the blank control group in the relational matching-to-sample task.

in a specific proportion, which differed in different groups, and the trials were sequenced in a pseudo-random manner. Of note, the proportion of AS and PS trials was 5:1 in the HEA group, and 1:1 in the LEA group.

#### *Relational Matching-to-Sample Task*

This task contained two conditions. In the relational reasoning condition, which reflected analogical reasoning ability, two pairs of objects (top and bottom) were presented simultaneously (Fig. 1B, left). Participants first had to deduce which dimension was incongruent between the top two objects (objects differed either in terms of shape

or texture), and then determine whether the bottom two objects differed along the same dimension. While in the perceptual matching condition, which reflected perception, three objects (two at the top and one at the bottom) were presented simultaneously (Fig. 1B, right). The participant had to determine if the bottom object matched either one of the top two objects along the specified dimension (either shape or texture). Due to the easiness of the task, if one’s accuracy of perception was <90% and less than the mean minus three SDs, the data were removed from the subsequent analysis as it was considered that the participant had not treated the experiment seriously.

The relational reasoning and perceptual matching conditions were presented in 24 alternating blocks (12 blocks per condition). The sequence of blocks was pseudo-random, and participants rested for at least 30 s after every 8 blocks. Each block was preceded by instructions that lasted 2000 ms. In the relational reasoning condition, the instructions stated, “Match Change Rule”, while in perceptual matching, they were either “Match Shape” or “Match Texture”, which had been translated into Chinese. Each trial began with the presentation of a fixation “+” for 500 ms, followed by the appearance of stimuli. The display lasted until the participant reacted, but no longer than 3500 and 2800 ms for the relational reasoning and perceptual matching conditions, respectively. After a response was made or the end of the maximum stimulus duration was reached, a fixation “+” was displayed again for the remaining trial duration. There were 8 trials per block in the relational reasoning condition (Fig. 1C, upper) and 10 trials per block in the perceptual matching condition (Fig. 1C, lower), which were randomly sequenced. In both conditions, participants were instructed to press “J” with the right index finger to indicate “yes” and “F” with the left counterpart for “no” [40].

## Experimental Procedure

First, continuous resting EEG was recorded for 5 min with eyes open. Then, participants performed the AS task following the instructions without practice. In addition, those in the HEA group were informed that the target would appear frequently at the opposite side of the cue, while participants in the LEA group were informed that the position of the appearing target would be unpredictable. Another 5 min of rest with eyes open and EEG recording was conducted when the task was accomplished. After a short break (~2–3 min), participants performed a short training session for familiarization with the stimuli before starting the final task (relational reasoning task or perceptual matching task). The entire experiment took ~1.5 h (Fig. 1D).

## EEG Processing and Statistical Analyses

### EEG Recording and Pre-processing

Continuous EEG was recorded from standard 64 in-cap Ag/AgCl electrodes, following the extended international 10–20 system (Brain Products GmbH, Munich, Germany) and two additional electrodes placed over both mastoids. Vertical and horizontal electrooculograms were recorded from below the left eye and outer canthus of the right eye, respectively. The cephalic electrode was used as an on-line reference. The impedance of all electrodes was maintained below 10 k $\Omega$  throughout the recording process. EEG and

electrooculogram data were continuously collected at a sampling rate of 500 Hz and on-line filtered with a 0.1–100 Hz bandpass filter.

Following data recording, EEG data were pre-processed using EEGLAB v13.0.1 [56] and MatLab R2014a (The MathWorks, Natick, USA). Offline EEG data were re-referenced to the mean of both mastoids. Data were filtered with a 1–30 Hz bandpass filter *via* a basic finite impulse response filter. Continuous EEG was epoched from –200 to 800 ms for the first cue markers in AS tasks and stimulus markers in relational matching-to-sample tasks and 2000 ms for the resting state. The baseline window was from –200 to 0 ms before the cue or stimulus. Independent components analysis was used across all four epochs to remove ocular artifacts. An epoch was automatically rejected whenever the voltage reached >100  $\mu$ V. At the end of these standard pre-processing steps, at least 90% of trials were retained per condition. Three topographic regions of interest (ROIs) were identified for future EEG analysis across the following electrode sites: anterior (Fz, F1–F4), midline (Cz, C1–C4), and posterior (Pz, P1–P4).

### ERP Analysis

Based on previous studies and grand mean mapping, cue-locked N2 in AS tasks, as well as stimulus-locked P2 and LPC in relational matching-to-sample tasks, were quantified as the mean amplitude ( $\mu$ V) during a 50-ms window (150–200 ms) [57], a 100-ms window (150–250 ms) [58], and a 400-ms window (400–800 ms) [59], respectively, in correct trials. The average peak latencies for perception and reasoning P2 in the HEA/LEA groups were 197.83/200.63 ms and 209.11/204.51 ms (anterior), 199.94/207.54 ms and 209.11/213.43 ms (midline), and 212.89/217.09 ms and 219.39/220.57 ms (posterior). For LPC, the counterparts were 603.11/582.40 ms and 628.44/622.17 ms (anterior), 600.56/574.34 ms and 596.56/567.37 ms (midline), and 549.79/523.20 ms and 562.56/538.34 ms (posterior). N2 amplitudes were analyzed using a two-way mixed design repeated-measures ANOVA with within-subject (AS vs PS) and between-subject (HEA vs LEA) factors. P2 and LPC amplitudes were analyzed using two-way mixed design repeated-measures ANOVA with within-subject (relational reasoning vs perceptual matching) and between-subject (HEA vs LEA) factors. Pearson correlation analyses were applied to explore the relationship between reasoning performance and ERP data, as well as the relationships between ERP data for different tasks. Of note, the *P*-values were corrected *via* the false discovery rate. Finally, a 5000 bootstrap re-sample analysis was used to calculate the bias-corrected 95% confidence intervals around the mediated and direct effects mediation analysis in Process v3.2 (<https://www.proce>



ssmacro.org) in SPSS22.0 (IBM, Armonk, New York, USA). An indirect effect was considered significant when the confidence interval did not include zero [60].

### Spectral Analysis

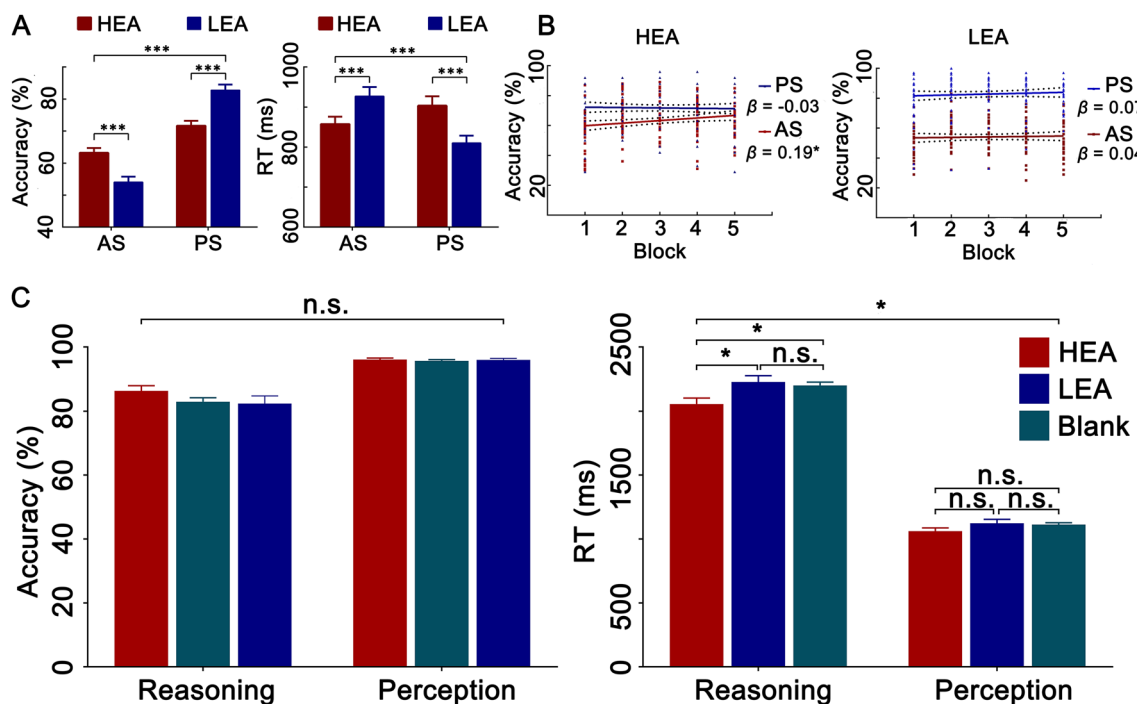
EEG data were down-sampled to 256 Hz after artificial removal and converted into time-frequency domain data using continuous wavelet transform (CWT) in Letswave software (<https://www.letswave.org/>). The cmor1-1.5 was chosen as the mother wavelet function for spectral analysis within alpha bands (8–14 Hz). This yielded CWT average power values for each EEG frequency band expressed in log units ( $\mu\text{V}^2/\text{Hz}$ ), as a measure of frequency density (activity) in all four recordings (pre-resting state HEA/LEA and post-resting state HEA/LEA). A two-way mixed design repeated-measures ANOVA with within-subject (pre vs post) and between-subject (HEA vs LEA) factors was applied. Then, we explored the relationship between the power values in significantly changed frequency bands in the post-resting state and during the reasoning tasks using Pearson's correlation.

## Results

### Behavioral Results

Two participants with perception accuracy scores below 90% (83% and 86%) and less than the mean  $-3\text{SDs}$  ( $-3.56\text{SDs}$  and  $-3.14\text{SDs}$ ) were excluded as it was considered that they were not taking the experiment seriously.

In AS tasks, a mixed-design repeated-measures ANOVA was applied to the accuracy rate and response time (trial type: AS vs PS  $\times$  group: HEA vs LEA) (Fig. 2A). For accuracy, the main effect of trial type was significant [ $F_{(1, 69)} = 218.73, P < 0.001, \eta^2_p = 0.76$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.26, P = 0.611$ ], and the two-way interaction between trial type and group was significant [ $F_{(1, 69)} = 64.88, P < 0.001, \eta^2_p = 0.49$ ]. Simple effect analysis indicated that the accuracy of AS trials in the HEA group ( $63.22\% \pm 1.48\%$ , mean  $\pm$  SEM) was significantly higher than in the LEA group ( $54.07\% \pm 1.72\%$ ) ( $P < 0.001$ ). In contrast, the accuracy of PS trials of the HEA group ( $71.69\% \pm 1.52\%$ ) was significantly lower than the LEA group ( $82.79\% \pm 1.74\%$ ) ( $P < 0.001$ ). For reaction time (RT), the main effect of trial type was significant [ $F_{(1, 69)} = 13.71, P < 0.001, \eta^2_p = 0.17$ ], the main effect of



**Fig. 2** The behavioral results of tasks. **A** The result of ANOVA for the accuracy (left) and reaction time (RT, right) in anti-saccade (AS) and pro-saccade (PS) trials between the experimental group (HEA,  $n=36$ ) and active control group (LEA,  $n=35$ ). **B** The result of unitary linear regression models for the accuracy in AS or PS trials changes as blocks go by in the HEA (left,  $n=36$ ) and LEA (right,

$n=35$ ) groups. **C** The accuracy (left) and reaction time (right) in the reasoning and perception condition in the HEA ( $n=36$ ), LEA ( $n=35$ ), and blank control (Blank) groups ( $n=100$ ). Error bars, SEM; n.s., not significant; \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ . HEA, high executive attention group; LEA, low executive attention group.

group was not significant [ $F_{(1, 69)} = 0.18, P = 0.671$ ], and the two-way interaction between trial type and group was significant [ $F_{(1, 69)} = 72.46, P < 0.001, \eta^2_p = 0.51$ ]. Simple effect analysis indicated that the HEA group completed AS trials significantly faster ( $857.56 \pm 18.56$  ms) than the LEA group ( $927.28 \pm 22.54$  ms) ( $P < 0.05$ ). In contrast, the HEA group completed PS trials significantly slower ( $903.67 \pm 23.19$  ms) than the LEA group ( $810.20 \pm 18.38$  ms) ( $P < 0.01$ ).

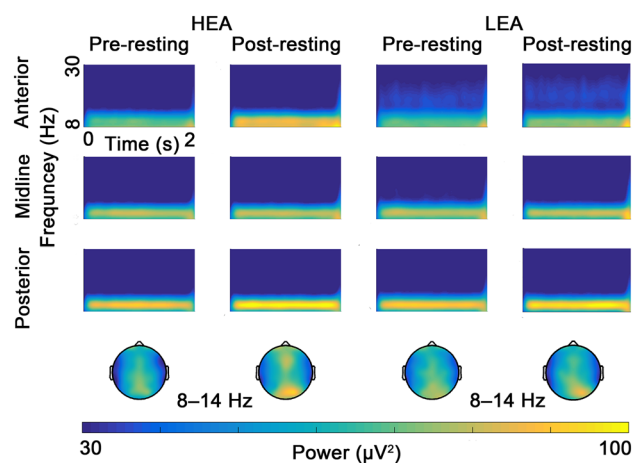
Unitary linear regression models between the accuracy of AS or PS trials and blocks were built to explore whether there was a practice effect. The results (Fig. 2B) showed that in the HEA group, the model referred to the accuracy of AS and blocks fitted well ( $\beta_{\text{normalized}} = 0.19, t = 2.58, P < 0.05, \Delta R^2 = 0.031$ ), but not in PS ( $\beta_{\text{normalized}} = -0.03, t = -0.34, P = 0.734, \Delta R^2 = -0.005$ ). The model fitting effects were both non-significant in the LEA group (AS:  $\beta_{\text{normalized}} = 0.04, t = 0.47, P = 0.637, \Delta R^2 = -0.004$ ; PS:  $\beta_{\text{normalized}} = 0.07, t = 0.88, P = 0.381, \Delta R^2 = -0.001$ ). Finally, the mean target duration across five blocks in the HEA group was significantly larger ( $94.33 \pm 1.88$  ms) than the LEA group ( $87.31 \pm 2.07$  ms) [ $t_{(69)} = 2.51, P < 0.05, d = 0.60$ ]. This revealed that although the HEA group performed better in AS trials, the task was still difficult (since the duration would reduce while the accuracy of block was  $>60\%$ ) and always beyond the capacity of the participants in the training, which helped to induce behavioral and neural plasticity.

In the relational matching-to-sample task, a mixed-design repeated-measure ANOVA was performed respectively on the accuracy rate and response time between the three groups across two conditions (condition: Reasoning vs Perception  $\times$  group: HEA vs LEA vs Blank). For accuracy, as shown in Fig. 2C (left), the main effect of condition was significant [ $F_{(1, 168)} = 141.44, P < 0.001, \eta^2_p = 0.46$ ], the main effect of group was not significant [ $F_{(2, 168)} = 1.12, P = 0.330$ ], and the two-way interaction between condition and group was not significant [ $F_{(2, 168)} = 1.11, P = 0.330$ ] (Reasoning<sub>HEA</sub>:  $86.34\% \pm 1.63\%$ ; Reasoning<sub>LEA</sub>:  $82.41\% \pm 2.37\%$ ; Reasoning<sub>Blank</sub>:  $82.96\% \pm 1.25\%$ ; Perception<sub>HEA</sub>:  $96.06\% \pm 0.46\%$ ; Perception<sub>LEA</sub>:  $95.95\% \pm 0.47\%$ ; Perception<sub>Blank</sub>:  $95.64\% \pm 0.40\%$ ). For RT, as shown in Fig. 2C (right), the main effect of condition was significant [ $F_{(1, 168)} = 3424.55, P < 0.001, \eta^2_p = 0.95$ ], the main effect of group was significant [ $F_{(2, 168)} = 4.16, P < 0.05$ ], and the two-way interaction between condition and group was also significant [ $F_{(2, 168)} = 3.16, P < 0.05$ ]. Simple effect analysis indicated that the main effect of group was significant in reasoning [ $F_{(2, 168)} = 4.67, P < 0.05$ ] but was not in perception [ $F_{(2, 168)} = 1.74, P = 0.179$ ]. *Post hoc* Bonferroni analysis indicated that the HEA group reacted significantly faster (Reasoning<sub>HEA</sub>:  $2056.47 \pm 46.41$  ms) than the LEA group (Reasoning<sub>LEA</sub>:  $2228.81 \pm 49.41$  ms;  $P < 0.05$ ) and the blank control group (Reasoning<sub>Blank</sub>:  $2201.81 \pm 25.81$  ms;

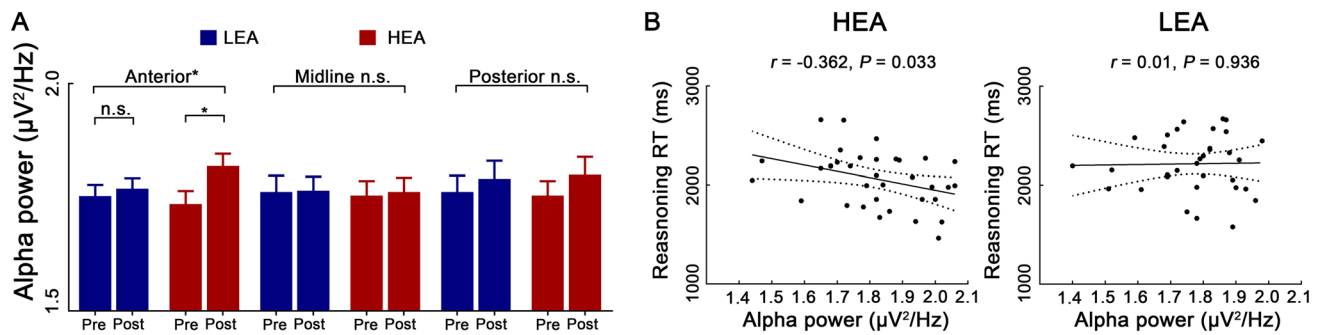
$P < 0.05$ ). There was no significant difference between the LEA group and the blank control group.

## Spectral Results

In the beginning, two participants were excluded for a high epoch reject rate ( $>70\%$ ) in pre-resting or post-resting state spectral analysis, but were recalled in the next ERP analysis as the epoch reject rate was low at that time ( $<5\%$ ). Fig. 3 shows the raw power of alpha (8–14 Hz) of pre- and post-resting states in the two groups. Then, a series of 2 (group: HEA vs LEA)  $\times$  2 (resting state: Pre vs Post) repeated-measures ANOVAs were conducted on the power for each ROI (anterior, midline, and posterior) after raw power values were transformed to  $\log_{10}$  units. For the alpha band (Fig. 4A), in anterior, the main effect of group was not significant [ $F_{(1, 67)} = 0.01, P = 0.921$ ], the main effect of resting state was significant [ $F_{(1, 67)} = 10.06, P < 0.01, \eta^2_p = 0.13$ ], and the interaction effect was significant [ $F_{(1, 67)} = 8.47, P < 0.01, \eta^2_p = 0.11$ ]. Simple effect analysis indicated that the power value of the post-resting state was significantly greater than the pre-resting state in the HEA group (pre:  $1.74 \pm 0.03 \mu\text{V}^2/\text{Hz}$ ; post:  $1.82 \pm 0.03 \mu\text{V}^2/\text{Hz}$ ), but they did not differ significantly from each other in the LEA group (pre:  $1.77 \pm 0.02 \mu\text{V}^2/\text{Hz}$ ; post:  $1.78 \pm 0.02 \mu\text{V}^2/\text{Hz}$ ). In midline, the main effect of group was not significant [ $F_{(1, 67)} = 0.46, P = 0.502$ ], the main effect of resting state was not significant [ $F_{(1, 67)} = 0.10, P = 0.753$ ], and the interaction effect was not significant [ $F_{(1, 67)} = 0.45, P = 0.503$ ]. In posterior, the main effect of group was not significant [ $F_{(1, 67)} = 0.58, P = 0.451$ ], the main effect of resting state was not significant [ $F_{(1, 67)} = 0.01, P = 0.947$ ], and the interaction



**Fig. 3** Spectral display in the resting state. Grand-average time-frequency representations in the anterior (Fz, F1–F4), midline (Cz, C1–C4), and posterior (Pz, P1–P4) in pre-resting and post-resting states in the experimental (HEA) and control (LEA) groups with the scalp topographies of the alpha band (8–14 Hz).



**Fig. 4** Changes in alpha power. **A** The result of ANOVA for the power in pre-resting and post-resting state alpha bands in the experimental (HEA,  $n=35$ ) and control (LEA,  $n=34$ ) groups in three

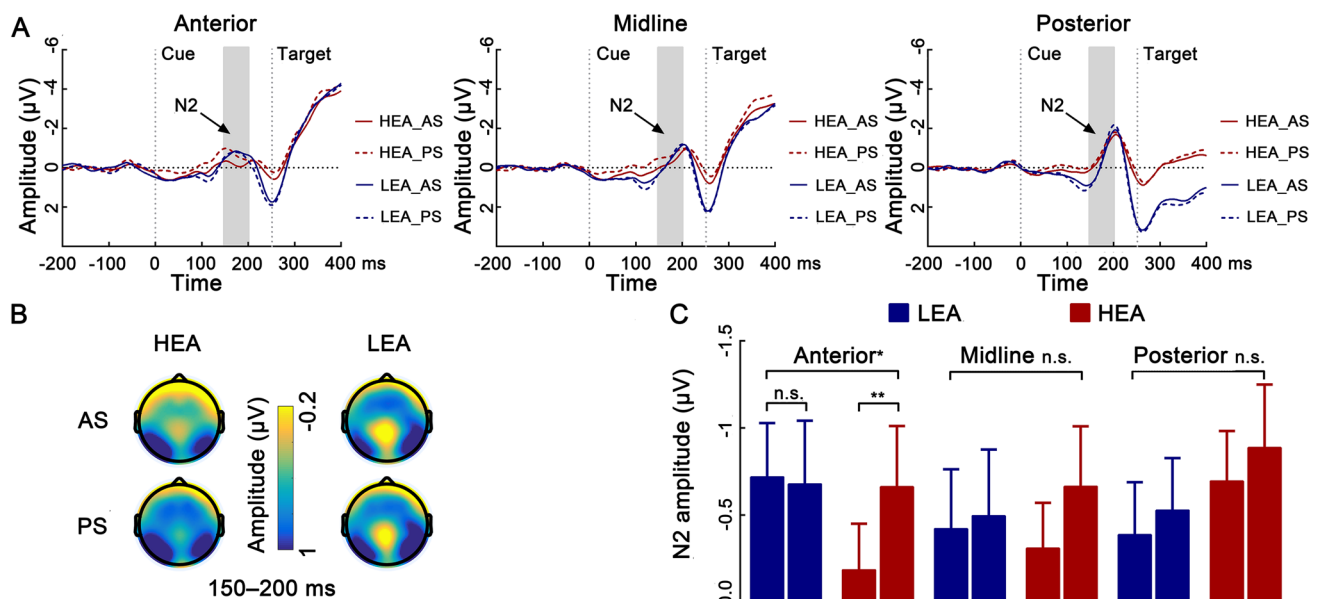
ROIs. Error bars, SEM; n.s., not significant;  $*P < 0.05$ . **B** Associations between alpha power of the post-resting state and reasoning reaction time in the HEA (right) and LEA (left) groups.

effect was not significant [ $F_{(1, 67)} = 0.69, P = 0.411$ ]. Then, we explored the association between reasoning reaction time and the anterior alpha power of the post-resting state. The results (Fig. 4B) revealed a significant negative correlation between reasoning reaction time and alpha power in the HEA group [ $r_{(35)} = -0.36, P < 0.05$ ], but not in the LEA group [ $r_{(34)} = 0.01, P = 0.936$ ].

## ERP Results

### N2

To evaluate differences between groups in N2 across the trial types, a series of 2 (group: HEA vs LEA)  $\times$  2 (trial type: AS vs PS) repeated-measures ANOVAs were conducted on N2 amplitude for each ROI (anterior, midline, and posterior) (Fig. 5A, B). In anterior, the main effect of trial type was marginally significant [ $F_{(1, 69)} = 3.84, P = 0.054, \eta_p^2 = 0.05$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.38, P = 0.538$ ], and the interaction effect was



**Fig. 5** N2 in anti-saccade tasks. **A** Event-related potential waveforms for the cue-locked N2 component in anti-saccade (AS) and pro-saccade (PS) trials in the two groups in three ROIs. Shaded regions represent the defined time windows of the N2 component. **B** Topographic distributions representing the average amplitude in the time

window (150–200 ms) across trial types from groups. **C** The result of ANOVA ( $n=71$ ) for the amplitude of AS and PS trials in three ROIs from groups for the N2 component in AS tasks. Error bars, SEM; n.s., not significant;  $*P < 0.05$ ,  $**P < 0.01$ .



significant [ $F_{(1, 69)} = 5.39$ ,  $P < 0.05$ ,  $\eta_p^2 = 0.07$ ]. Simple effect analysis indicated that N2 amplitudes of AS trials in the HEA group ( $-0.19 \pm 0.27 \mu\text{V}$ ) were significantly more positive than PS trials ( $-0.66 \pm 0.35 \mu\text{V}$ ;  $P < 0.01$ ); however, in the LEA group, the N2 amplitudes were not significantly different (AS:  $-0.72 \pm 0.31 \mu\text{V}$ ; PS:  $-0.68 \pm 0.36 \mu\text{V}$ ). In midline, the main effect of trial type was marginally significant [ $F_{(1, 69)} = 3.43$ ,  $P = 0.068$ ,  $\eta_p^2 = 0.05$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.004$ ,  $P = 0.951$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 1.47$ ,  $P = 0.229$ ]. In posterior, the main effect of trial type was not significant [ $F_{(1, 69)} = 1.89$ ,  $P = 0.174$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.61$ ,  $P = 0.438$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.04$ ,  $P = 0.834$ ] (Fig. 5C).

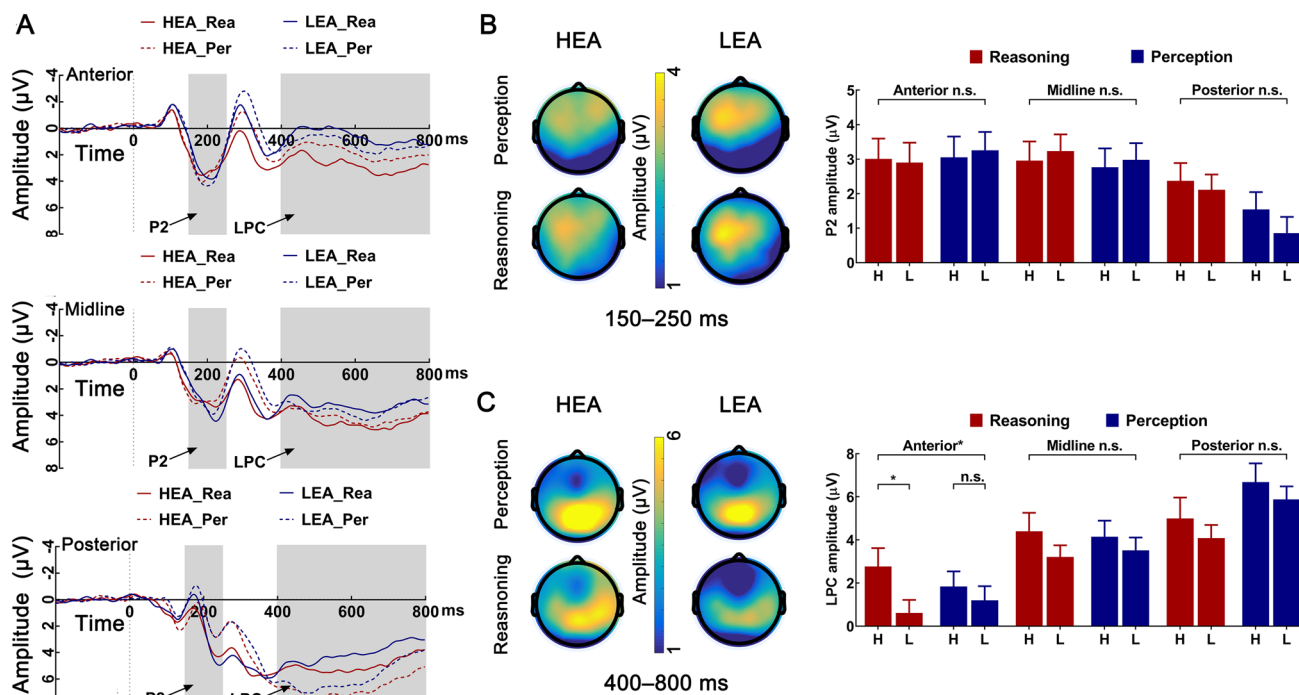
## P2

Figure 6A shows the waveforms between  $-200$  and  $800$  ms for each group in different ROIs for each condition. A series of 2 (group: HEA vs LEA)  $\times$  2 (condition: relational reasoning vs perceptual matching) repeated-measures ANOVAs were conducted on P2 amplitude for each ROI. As Fig. 6B shows, in anterior, the main effect of condition was

not significant [ $F_{(1, 69)} = 1.15$ ,  $P = 0.288$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.004$ ,  $P = 0.951$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.69$ ,  $P = 0.409$ ]. In midline, the main effect of condition was not significant [ $F_{(1, 69)} = 1.50$ ,  $P = 0.225$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.12$ ,  $P = 0.728$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.03$ ,  $P = 0.866$ ]. In posterior, the main effect of condition was significant [ $F_{(1, 69)} = 23.48$ ,  $P < 0.001$ ,  $\eta_p^2 = 0.25$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.52$ ,  $P = 0.473$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.97$ ,  $P = 0.329$ ].

## LPC

Similarly, a series of 2 (group: HEA vs LEA)  $\times$  2 (condition: relational reasoning vs perceptual matching) repeated-measures ANOVAs were conducted on LPC amplitude for each ROI. As Fig. 6C shows, in anterior, the main effect of condition was not significant [ $F_{(1, 69)} = 0.27$ ,  $P = 0.605$ ], the main effect of group was not significant [ $F_{(1, 69)} = 2.16$ ,  $P = 0.146$ ], and the interaction effect was significant [ $F_{(1, 69)} = 5.25$ ,  $P < 0.05$ ,  $\eta_p^2 = 0.07$ ]. Simple effect analysis indicated that LPC



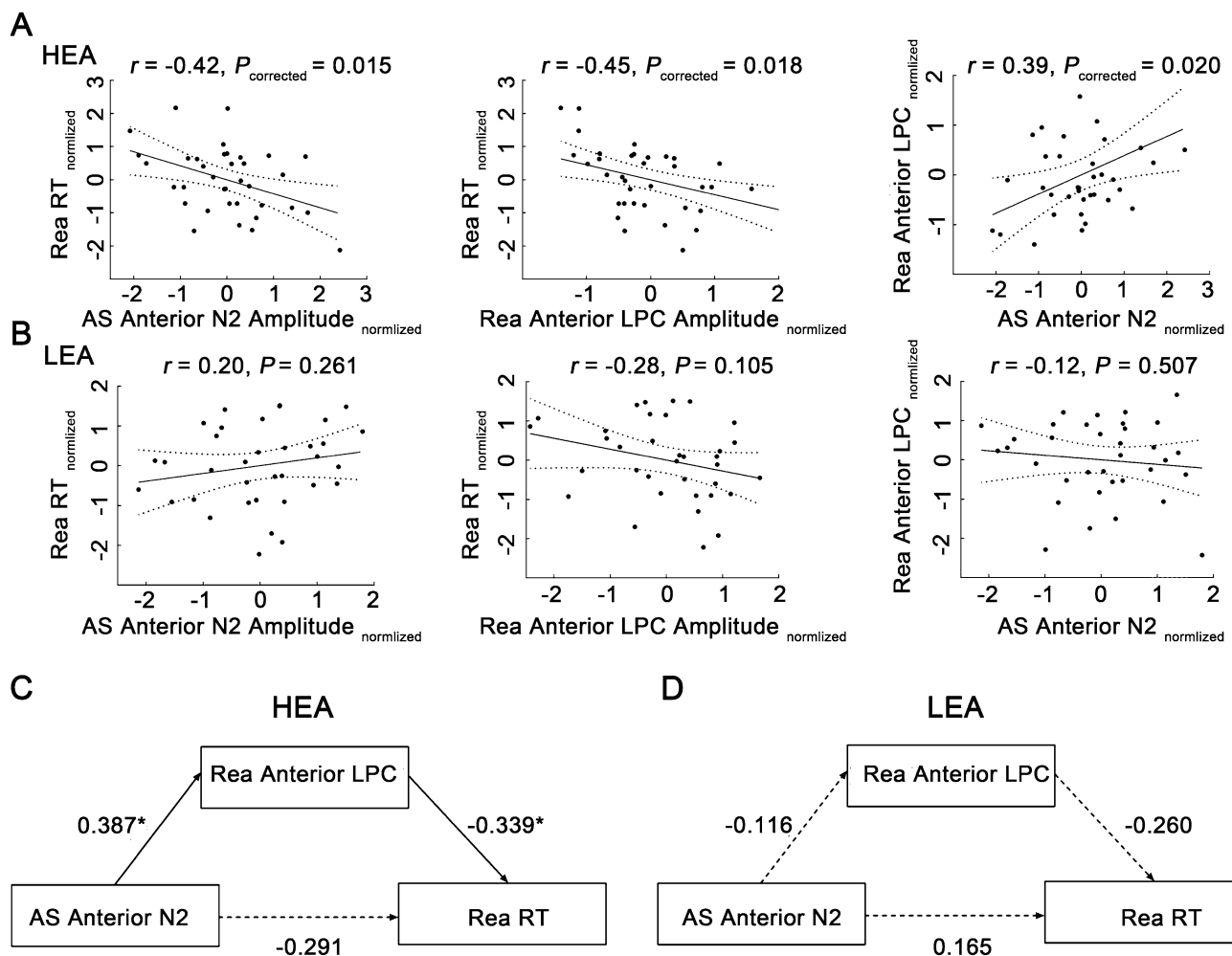
**Fig. 6** P2 and LPC in the relational matching-to-sample task. **A** Event-related potential waveforms for the stimulus-locked P2 and late positive component (LPC) in perception (Per) and reasoning (Rea) conditions in the two groups in three ROIs. The shaded regions represent the defined time windows of the P2 component and LPC. **B**, **C** Topographic distributions of average amplitude in the time win-

dows (**B**, 150–250 ms; **C**, 400–800 ms) across conditions between groups (left). The result of ANOVA ( $n = 71$ ) for the amplitude of the P2 (**B**) and LPC (**C**) amplitudes for reasoning and perception in three ROIs between groups (right). Error bars, SEM; n.s., not significant; \* $P < 0.05$ . H, the HEA group; L, the LEA group.

amplitudes of reasoning in the HEA group ( $2.77 \pm 0.85 \mu\text{V}$ ) were significantly more positive than the LEA group ( $0.61 \pm 0.61 \mu\text{V}$ ) ( $P < 0.05$ ); however, LPC amplitudes of perception were not significantly different between the two groups. In midline, the main effect of condition was not significant [ $F_{(1, 69)} = 0.01$ ,  $P = 0.936$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.96$ ,  $P = 0.330$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.73$ ,  $P = 0.396$ ]. In posterior, the main effect of condition was significant [ $F_{(1, 69)} = 29.45$ ,  $P < 0.001$ ,  $\eta^2_p = 0.30$ ], the main effect of group was not significant [ $F_{(1, 69)} = 0.65$ ,  $P = 0.422$ ], and the interaction effect was not significant [ $F_{(1, 69)} = 0.03$ ,  $P = 0.863$ ].

### Association Between AS Anterior N2 and Relational Reasoning Reaction Time

At the between-group level, we found that the AS anterior N2 amplitudes decreased in the HEA group, accompanied by increased anterior LPC amplitudes in the sequential reasoning task. This suggested that a decreased AS anterior might lead to an increased anterior LPC, which finally causes a better reasoning performance that is reflected by a faster RT. If so, at the individual level, the anterior LPC might mediate the relationship between the AS anterior N2 and reasoning RT. To test this hypothesis, we first conducted multiple correlation analyses with  $P$ -values corrected by the false discovery rate. The results showed that in the HEA group (Fig. 7A), the AS anterior N2 and reasoning anterior



**Fig. 7** Correlation between physiology and psychology in the reasoning condition. **A, B** Pearson's correlation analysis reveals the association between reasoning reaction time (Rea RT), anti-saccade (AS) anterior N2 amplitudes, and reasoning anterior late positive component (LPC) amplitudes in the experimental (HEA,  $n = 36$ ) (upper) and control (LEA,  $n = 35$ ) (lower) groups. **C, D** Mediation analysis

reveals that reasoning anterior LPC amplitudes fully mediate the association between AS anterior N2 and Rea RT in the HEA group (left,  $n = 36$ ) which is not found in the LEA group (right,  $n = 35$ ). Note that the number beside the arrow indicates the normalized regression coefficients and the significance level.  $^*P < 0.05$ .

LPC amplitudes were both significantly and negatively correlated with the reasoning RT [ $r_{(36)} = -0.42$ ,  $P_{\text{corrected}} < 0.05$ ;  $r_{(36)} = -0.45$ ,  $P_{\text{corrected}} < 0.05$ ], and they were significantly and positively correlated [ $r_{(36)} = 0.39$ ,  $P_{\text{corrected}} < 0.05$ ]. However, in the LEA group (Fig. 7B), no significant relationship was found, although the reasoning anterior LPC was also negatively correlated with reasoning RT [ $r_{(35)} = -0.28$ ,  $P = 0.105$ ]. Then, mediation analyses were conducted based on the correlation outcomes. In the HEA group (Fig. 7C), the normalized regression coefficient between AS anterior N2 and reasoning RT was marginally significant, but the normalized regression coefficient between reasoning anterior LPC and AS anterior N2 and reasoning RT was significant. The normalized indirect effect was  $-0.131$  and bootstrapping procedures indicated that the 95% confidence interval (CI) of the normalized indirect effect ranged from  $-0.285$  to  $-0.003$ , suggesting a statistically significant indirect effect. Nevertheless, in the LEA group (Fig. 7D), no significant indirect effect was found (indirect Effect<sub>normalized</sub> =  $0.030$ , 95% CI:  $[-0.98, 0.136]$ ).

## Discussion

Consistent with our predictions, the HEA group reacted significantly faster than the LEA group and the blank control group in the reasoning condition, but this was not so in the perception condition after executive attention intervention. At the neural level, only the HEA group showed significantly elevated changes in anterior alpha activity, which was related to the subsequent reasoning RT. Furthermore, the LPC amplitude in reasoning was more positive in the HEA group than the LEA group, but there was no difference between the two groups in perception. Importantly, the LPC amplitude mediated the relationship between the N2 amplitude of AS trials in the AS task and reasoning RT. These results provide causal data that may help to understand why executive attention interventions improve analogical reasoning.

Behaviorally, the HEA group showed a reversed response pattern compared to the LEA group in AS tasks (Fig. 2A), and the accuracy of AS trials in the HEA group continuously increased with time (Fig. 2B). When the proportion of AS trials was higher than PS, executive attention is recruited more intensively [61, 62] and proactive control can be improved [63] when the cue is predictive, and these results suggested that the executive attention intervention in the HEA group was successful. Namely, the HEA group performed better in inhibitory control and voluntary attentional shift than the LEA group. Then the main finding was that the HEA group responded significantly faster than the LEA group and the blank control group only in the reasoning condition after executive attention intervention (Fig. 2C, right). This result showed that executive attention intervention

affected analogical reasoning rather than the lower-level perception. On the one hand, executive attention overlaps with inhibitory control [64], especially the AS task, which is the typical measurement for this cognitive function [65]. Theoretically, inhibitory control is the key psychological mechanism of analogical reasoning which helps to prevent task-irrelevant information from competing for processing resources [66–71]. A recent study reported that, compared to the neutral condition, participants react faster while solving analogical reasoning problems after finishing a Stroop trial, providing causal evidence that inhibitory control plays a central role in analogical reasoning [72]. However, inhibitory control may not be the only factor contributing to enhanced analogical reasoning performance as recent studies have revealed that inhibitory control fails to predict analogical reasoning performance when age is controlled [73]. On the other hand, eye-tracking studies emphasized that attention needs to be correctly distributed on the different objects to successfully solve analogical reasoning problems [51, 52]. In other words, only when the attention of the individual shifts voluntarily and rapidly between different objects for feature comparison, can the relationship between these objects likely be successfully extracted, and then the correct answer would be generated. In all, the core cognitive mechanism of executive attention intervention may lie in enhanced inhibitory control and voluntary attentional shift, which certainly helps the information process.

Importantly, the different neural activity patterns in the two groups across conditions provided deep insight into the mechanism by which executive attention intervention improves analogical reasoning. First, we found that only anterior resting alpha activity was significantly enhanced after AS tasks in the HEA group (Figs 3, 4A). Previous studies mostly found that decreased alpha is related to increased arousal and selective attention while individuals are conducting cognitive tasks [74, 75]. However, the resting stage was characterized by low arousal where individuals were instructed to control their attention on the fixation instead of other mental activities to maintain focus. A previous study also found that resting anterior alpha activity improves after ~12 min of a single session of attention training [36]. Since alpha oscillations are generated by fronto-parietal executive control networks [76], thought to reflect the engagement of executive attention [77] and closely related to the AS task process [78], the increased alpha power suggested that a high proportion of AS tasks could improve executive attention level, even after only one session. Moreover, the HEA group showed that enhanced anterior power was negatively correlated with the reasoning RT (Fig. 4B). Since participants did need to block distractions from the environment to focus on the fixation in the resting stage and alpha activity is related to the ability to suppress distractors or irrelevant information [31, 33], this suggested that individuals in the HEA

group with higher alpha power gained in executive attention intervention might exert better resolution of interference, which is one of the core abilities required in the analogical reasoning task [79]. Moreover, the LEA group did not benefit from the executive attention task and it was costly both in terms of time and cost in preparing the EEG experiment and finishing the executive attention task, the individuals in alpha power might thus not reflect the executive attention level as intensively as the HEA group, which then caused a weak correlation.

Second, we found that the anterior N2 of AS was smaller than that of PS in the HEA group, but they were comparable in the LEA group (Fig. 5), which was different from the previous study that found the AS task with indicated cues results in a larger positive cue-locked N2 in PS than in AS [26]. Since increased N2 implied that AS is a conflict [80], while N2 amplitude is reduced due to the adaption to interference [46] and an increased executive attention level [47], here, smaller amplitudes reflected a stronger ability to resolve the interference. In addition, the cue was not predictive in the LEA group, resulting in no difference in patterns during preparation for AS or PS. This result was reasonable and indicated that the resolution of interference in the HEA group had been improved during executive attention intervention. Third, P2 in all ROIs showed no significant difference in task condition, group, and interaction (Fig. 6A, B). P2 is related to perceptual processing [43], reflecting the early stages of target discrimination and identification [42]. Our results again supported the notion that the contribution of executive attention to analogical reasoning ability was not *via* enhanced perception, which also excluded the existence of a behavioral ceiling effect. Finally, reasoning elicited significantly larger anterior LPC amplitudes in the HEA group than in the LEA group, while no significant difference was found between the two groups in perception (Fig. 6A, C). Generally, the pattern of the ERP results in a cognitive intervention is an increased ERP component related to the cognitive process [48]. LPC amplitude may indicate a functional and appropriate response to external stimuli [81, 82]. Specifically, a previous study proposed that frontal LPC in analogical reasoning tasks is correlated with relation integration [30]. Since relation integration is the key to the analogical reasoning process [79], the results most likely reflected the increased ability of the participants to focus on the relational information, which finally led to better analogical reasoning performance. Compared to perception matching, the results also supported the idea that executive attention intervention directly enhances higher-order cognitive processing rather than lower-level sensory ability.

More importantly, the anterior LPC amplitudes significantly mediated the relationship between the anterior N2 of AS trials and RT in reasoning in the HEA group, but not in the LEA group (Fig. 7). As discussed above, a smaller N2 of AS

trials indicated a higher ability to resolve interference in the HEA group. In the HEA group, participants needed more voluntary attentional shift as most of the trials were AS in the AS tasks, indicating that those who were better at voluntary attention control would also concentrate more on relation integration, which then helped solve analogical reasoning problems. This is in line with the unbalanced attentional focus hypothesis proposing that relation integration needs proper attention control, which finally helps solve analogical reasoning problems [54]. Another previous study found a partial overlap between the neural correlates of resolution interference and relation integration [79]. Therefore, this mediator effect revealed why executive attention intervention could improve analogical reasoning from the perspective of individual differences. In addition, Bayesian learning accounts assume that individuals have naive priors that are updated due to external informative cues, yielding posterior distribution modulations that have been thought to reflect the development of cognitive abilities [83, 84]. Recently, a large-scale eye-movement study found that the mean saccade amplitude is larger when a task is solved than when it is not, in the fluid intelligence test [23], indicating that a voluntary shift in attention control can contribute to the successful solving of a reasoning task. This suggests that executive attention intervention could be a cue predicting future contexts where individuals would extract the relationship needed for intense voluntary attention control to shift from A and B during the analogical reasoning task. It also helps to explain why the LEA group did not obtain enhanced analogical reasoning, namely, less experience in voluntary attention control caused these participants to fail to predict the future context, which ultimately failed to facilitate improvements in analogical reasoning. More importantly, to a certain extent, this could explain why previous cognitive interventions applied in other executive attention tasks (go/no-go and stop signal) do not transfer to reasoning tasks [85]. Since both tasks did not require the participant to shift their attention voluntarily, the training tasks could not act as cues to predict future tasks. Although executive attention is a general ability, it is specially required in different tasks [11] and the AS task is believed to dominate latent factors of executive attention [18, 19]. This helps to explain why other executive attention task training failed to facilitate an improvement in reasoning, and AS tasks succeeded. However, more neural research is needed to explore why different executive attention training tasks have different transfer effects.

Collectively, all significant neural results were found in the anterior regions of the brain, suggesting that executive attention and analogical reasoning share some neural mechanisms. A previous study found that visual analogical reasoning is positively correlated with the activation level of the prefrontal executive network [86]. More broadly, research has shown that various tasks commonly used to measure fluid intelligence activate a specific frontal network involved



in the control of attention and behavior, including the mid-dorsolateral, mid-ventrolateral, and dorsal anterior cingulate cortex [15, 87]. Moreover, a recent study also demonstrated the co-evolution of executive attention and intelligence. They found that several regions of the frontal cortex, particularly the anterior cingulate cortex, which shows strong expansion in both development and evolution, is the key node of the executive attention network and is related to individual fluid intelligence [88]. This evidence suggested that executive attention is at the heart of intelligence and functions to regulate the flow of information in the mental working space of individuals [89]. So-called “general intelligence” probably refers to the ability to actively control and allocate attention. This was why analogical reasoning performance, the core ability in fluid intelligence, improved after executive attention intervention in this study.

In summary, we determined whether executive attention intervention could improve analogical reasoning, the core component of fluid intelligence, and aimed to elucidate the neural mechanisms underlying this improvement. We provided direct evidence that effective intervention (indexed by greater post-resting alpha power and smaller N2 effects of AS) enhanced analogical reasoning (indexed by shorter reasoning reaction time). The evidence showed that, compared to perception, executive attention mainly contributed to analogical reasoning *via* enhancing relation integration instead of perceptual ability (indexed by undifferentiated P2 and perception LPC, but larger reasoning LPC). Moreover, the mechanism underlying improvements was corroborated by significant mediating effects and consistent brain regions across the two tasks. Although more research is needed to replicate and extend our results, our results imply that executive attention intervention also improves the performance in more complex reasoning tasks, such as the Raven’s Progressive Matrices [90]. Given that nowadays cognitive training is quite controversial and some meta-analyses have denied vast transfer effects after working memory training [91, 92], this study not only promotes our understanding of the psychological mechanism of analogical reasoning, but also provides a new potentially promising training task for future cognitive training aimed at fluid intelligence improvement under a strict experimental design that includes baseline measurements and expectation control [93].

**Acknowledgements** This work was supported by grants from the National Natural Science Foundation of China (32171040 and 31900803), and the Graduate Research Innovation Project of Chongqing (CYS20094).

**Conflict of interest** The authors declare no potential conflict of interest.

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